Ian Leung

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Experience

Roblox **Jun 2026 – Aug 2026**
Software Engineer Intern, Network Infrastructure San Mateo, CA

- Worked alongside network engineers to automate network deployments, pushing to **5000+ production devices** in global datacenters with a GitHub Actions CI/CD pipeline equipped with staged expansion and health checks
- Orchestrated Temporal Workflows with LLM analysis & Slack alerting for safe, autonomous rollout of configuration changes, saving **400+ engineering hours** annually and identifying issues on 20+ network devices
- Identified and fixed a resource leak in an internal network service, recovering 7% CPU utilization across devices

WQC **Jan 2026 – Present**
Software Engineer, Core Services Waterloo, ON

- Built and deployed 2v2bluff, a real-time multiplayer Rust/WebSocket card game for four concurrent players
- Engineered portal.waterlooquantclub.com, a full-stack club recruitment & event portal hosted on AWS for **1200+ users**

Tells.dev **Apr 2026 – Jun 2026**
Co-Founder Waterloo, ON

- Built stealth browser infrastructure enabling AI agents to survive production bot detection in autonomous web sessions
- Scaled startup from 0 to 14 paying customers in its first month, amassing **50k+ minutes of session time** and generating interest from angel investors & VC firms

Sunnybrook Research Institute **Jul 2025 – Aug 2025**
Software Research Intern, Process Automation Toronto, ON

- Built an event tracing, playback, and logging tool that records and simulates clinician workflows in Focused Ultrasound software, surfacing 5+ defects and edge cases in real usage patterns
- Automated regression testing by generating synthetic JSON event flows and validating results against recorded baselines, **cutting testing time by 91%**

Projects

OuterwoRLd: Reinforcement Learning & 2D Physics Game | C++, Python, OpenGL, PyTorch, TensorBoard

- Built a 2D spaceflight physics game in C++ and OpenGL, with procedural level generation and gravity-based obstacles
- Implemented Proximal Policy Optimization (PPO) from scratch in PyTorch, training a reinforcement learning agent that reaches a **96% success rate on newly-generated levels**, up from 3% with the initial agent
- Diagnosed a credit-assignment failure (0.3 s of effective lookahead time) and fixed it with an 8-step action repeat, teaching the agent to dodge obstacles and raising success on obstacle-avoidance levels from 1% to 100%

Trust No Ghost: 3D Maze Horror Game | Unity, C#, Itch.io

- Shipped Trust No Ghost to **60+ players** at the UW Game Jam Fall 2025, placing in the **top 10** most-rated games
- Programmed ghost AI in C# with NavMesh pathfinding, wander/chase states, and proximity-based heartbeat audio

Jevis: Browser-use Agent Harness (Hack the North 2026 Winner) | Python, Playwright, JEV, OpenAI

- Engineered an agentic harness for browser-use using the JEV System One Model, automating browser tasks through a web interface and **cutting token cost by 250x** compared to GPT-6 Astra
- Sped up queries on Walmart from **5 minutes to 44 seconds** using a DOM-based approach and orchestrator models

Skills

Languages	Python, C++, Java, C#, Rust, TypeScript, SQL, Bash
Technologies	Unity, OpenGL, Temporal, gRPC, REST, PyTorch, Pandas, NumPy, OpenCV, PostgreSQL, LLMs, React
Tools	Linux, Git, Docker, AWS, GitHub Actions, Grafana, TensorBoard, Android Studio, Hermes Agent

Education

University of Waterloo **Sep 2025 – May 2029**
Bachelor of Computer Science & Finance GPA: 3.8/4.0

- President's Scholarship of Distinction ($5000), Undergraduate Achievement Award ($1500)
- Relevant Coursework: Object-Oriented Programming (Advanced), Computer Architecture (ARM), Linear Algebra

Competitions

Galois Math Contest Honor Roll · Canadian Computing Competition Distinction · ICDC International Finalist